RBT205 – Mechanisms and Materials

5.1 Assembling and Running a Set of Four Gears

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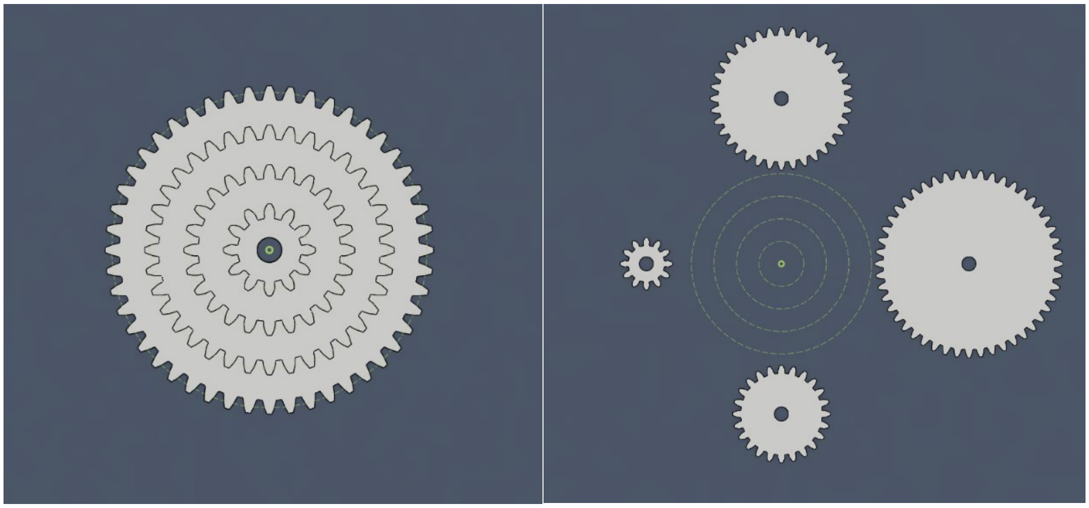
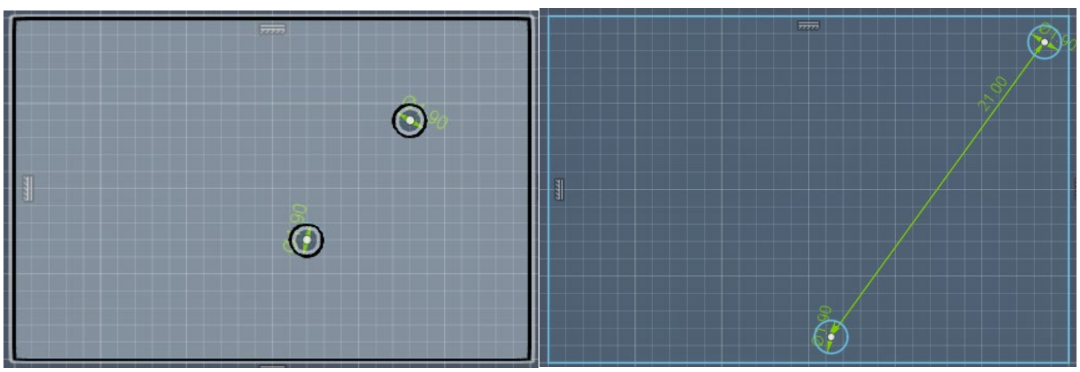
June 22, 2026

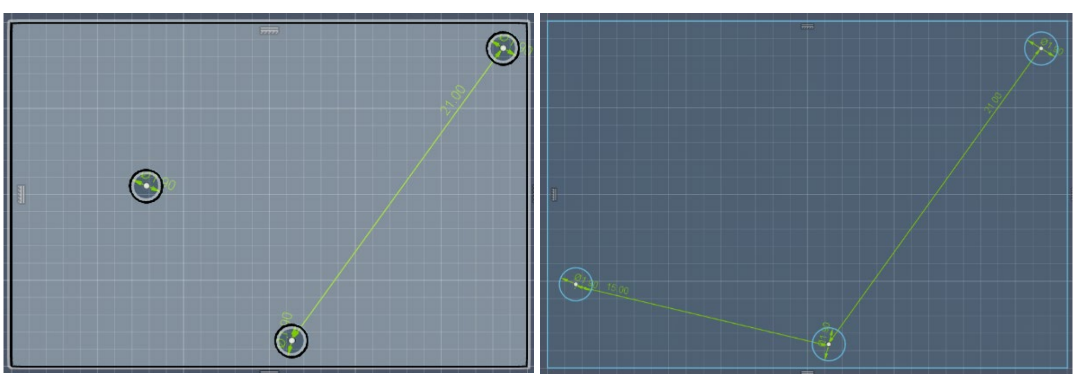
1. Created 4 gears with 48,36,24,12 teeth accordingly each with a 1.9-inch diameter holes.



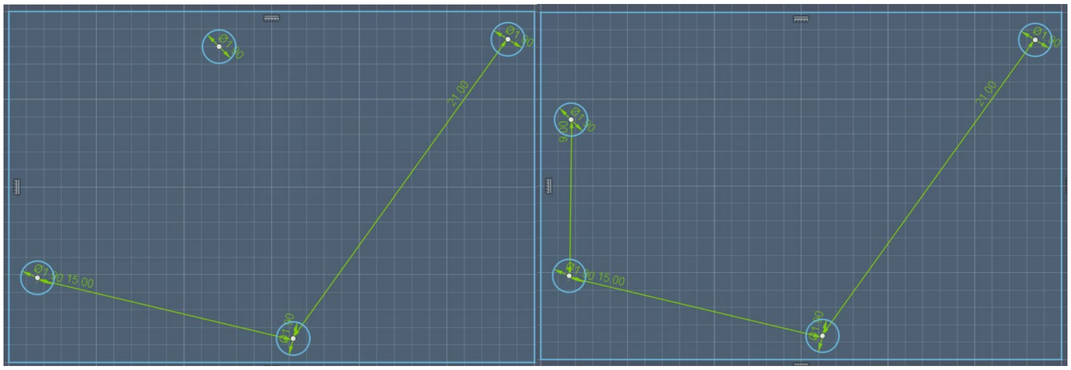
2. Created plate for mounting gears with holes measuring 1.9-inch diameter. Measurement of 4th and 3rd radius equals 21 inches. Set the holes to 21 inches apart.



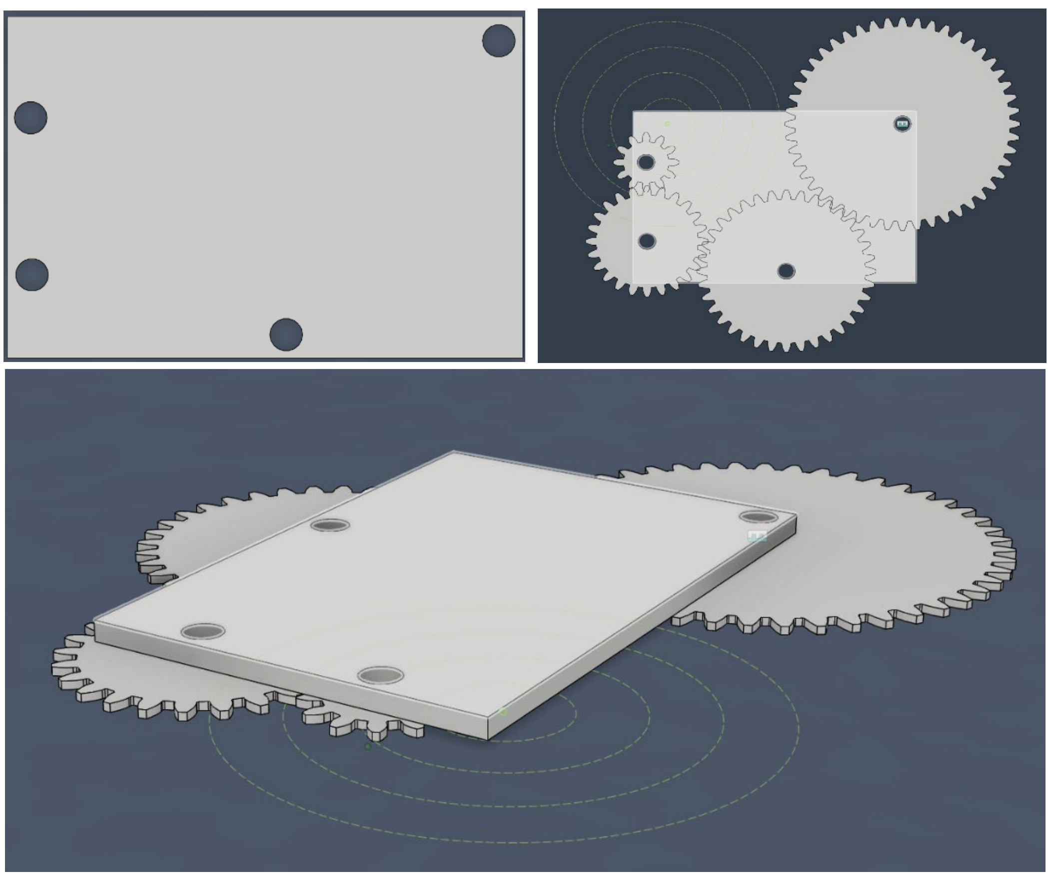
3. Repeated with the 3rd and 2nd radius; equal to 15 inches. Set the new hole 15 inches apart.



4. Repeated a final time with the 2nd and 1st radius; equal to 9 inches.



5. Made sketch into components. The overall gear ratio would be 1:4; with gears ranging from 12-48 the largest should turn once for 4 rotations of the smaller gear.



6. Link to: [Working Gear Set]